Mutation Domino

Confidence-aware missense-variant structural-context report

Analysis dfc347a6-f325-4604-9e3c-d7d823c3fe3f - generated 2026-10-25T18:14:24.773153Z - export 2026-10-25T18:20:46Z

RESEARCH AND EDUCATION ONLY. This report describes wild-type structural context and coarse physicochemical descriptors. It does not model the mutant structure, does not establish changed contacts or stability, and is not a pathogenicity or clinical prediction.

1. Input and mapping

Field	Value
UniProt accession	P04637
Protein	Cellular tumor antigen p53
Organism	Homo sapiens
Substitution	R175H
Canonical position	175
Structure chain / residue	A / 175
Residue	ARG (ARG)
Mapping status	mapped
Alignment coverage / identity	1.0 / 393 of 393 aligned columns

2. Data provenance

Field	Value
Analysis mode	live
Protein source	UniProt REST
Protein URL	https://rest.uniprot.org/uniprotkb/P04637.json
Protein retrieved	2026-10-25T17:56:34Z
Structure source	AlphaFold DB
Structure URL	https://alphafold.ebi.ac.uk/files/AF-P04637-F1-model_v6.cif
Structure model version	v6
Structure chain	A
Structure retrieved	2026-10-25T17:56:36Z
SHA-256	4332bff49301d04434b6e3b43c439aa5bd8238760e88c7142f7d0c0851a68e5d
Predicted model	yes
Algorithm version	1.0.0
Schema version	1.0.0
Parameters	{'ca_radius_angstrom': 8.0, 'heavy_atom_radius_angstrom': 5.0}

3. Confidence

pLDDT estimates local confidence in the wild-type predicted structure. It is not a prediction of mutation effect, stability, or pathogenicity.

Field	Value
Source	AlphaFold pLDDT
Availability	available
Per-residue value	96.62
Band	Very high (pLDDT > 90)
Note	For AlphaFold models the per-residue CA B-factor field stores pLDDT.

4. Residue chemistry (coarse descriptors)

Descriptor	Wild type	Mutant	Delta
Category	basic	basic	same
Charge class	positive	positive	same
Aromatic	False	True	changed
Hydropathy (Kyte-Doolittle)	-4.5	-3.2	1.3
Residue mass (Da)	156.19	137.14	-19.05
Side-chain heavy atoms	7	6	-1

Static coarse descriptors: Kyte & Doolittle (1982) hydropathy; average residue masses; manual side-chain heavy-atom counts. Not energy, stability, or function predictions.

5. Wild-type structural neighborhood

- In the supplied predicted wild-type structure (AlphaFold DB, chain A), residue ARG175 (canonical 175) of P04637 has 14 CA neighbour(s) within 8 Å.

- Its residue has 16 other residue(s) with at least one heavy-atom pair within 5 Å.

- The substitution Cellular tumor antigen p53 R175H is analysed only for wild-type structural context and coarse physicochemical property differences.

CA neighbors within 8.0 Angstrom: 14. Heavy-atom contacts within 5.0 Angstrom: 16.

Closest CA neighbors (max 12 shown, closest first)

Residue	Canonical	One-letter	Category	CA distance (A)
CYS176	176	C	polar	3.81
ARG174	174	R	basic	3.82
GLY245	245	G	special	5.03
LEU194	194	L	hydrophobic	5.5
CYS238	238	C	polar	5.62
GLN192	192	Q	polar	5.78
GLU180	180	E	acidic	5.84
HIS179	179	H	basic	6.29
PRO177	177	P	special	6.53
HIS193	193	H	basic	6.63
MET237	237	M	hydrophobic	7.01
VAL173	173	V	hydrophobic	7.18

Closest heavy-atom contacts (max 12 shown, closest first)

Residue	Canonical	One-letter	Category	Min heavy-atom distance (A)
CYS176	176	C	polar	1.33
ARG174	174	R	basic	1.34
PRO191	191	P	special	2.8
MET237	237	M	hydrophobic	2.86
HIS179	179	H	basic	2.94
HIS193	193	H	basic	3.3
GLN192	192	Q	polar	3.68
GLU180	180	E	acidic	3.86
PRO177	177	P	special	3.9
LEU194	194	L	hydrophobic	3.91
ARG196	196	R	basic	4.03
CYS238	238	C	polar	4.11

6. Interpretation and cautions

- Residue category is unchanged.

- Charge class is unchanged.

- Hydropathy delta (Kyte-Doolittle, mutant minus wild type): +1.30.

CAUTION: This report does not establish a changed contact network, stability change, binding disruption, pathogenicity, or any observed mutant effect.

CAUTION: The 3D coordinates are the wild-type reference model. No mutant structure was modelled or predicted.

CAUTION: CA neighbours and heavy-atom contacts are geometric descriptors, not hydrogen bonds, salt bridges, or energetic couplings.

CAUTION: pLDDT, when shown, is confidence in the wild-type predicted local structure, not mutation-outcome confidence.

CAUTION: Research and education only. Not for clinical decision-making.

Method summary: canonical UniProt sequence was aligned globally to the selected structure chain; the requested resident was required to map uniquely with matching residue identity. Distances were computed on the supplied wild-type coordinates using heavy (non-hydrogen) atoms for contacts. No mutant coordinates were generated.

Cite data sources: UniProt (https://www.uniprot.org) and AlphaFold Protein Structure Database (https://alphafold.ebi.ac.uk). Model and license terms remain those of the original providers.

Gene: TP53